Mathematics Method for Physics–Lecture 1

Complex Numbers and the Complex Plane

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Start up saying

1 Fundamentals and Representations of Complex Numbers

1.1 Algebraic (Cartesian) Form

Definition 1 (Complex number)**.** A **complex number** z is defined as an ordered pair of real numbers $(x, y) \in \mathbb{R}^2$, expressed in the algebraic form

$$z \coloneqq x + \mathrm{i}y, \quad x, y \in \mathbb{R}, \tag{1}$$

where

- i denotes the imaginary unit defined by $\mathrm{i}^2 \coloneqq -1$;
- $x \coloneqq \mathrm{Re}(z)$ is the **real part** of z;
- $y \coloneqq \mathrm{Im}(z)$ is the **imaginary part** of z.

1.1.1 Geometric Correspondence (The Complex Plane / Argand Plane)

There exists a canonical bijective correspondence among three entities:

1. The complex number $z = x + \mathrm{i}y \in \mathbb{C}$.
2. The point $P(x, y)$ in the Cartesian plane $\mathbb{R}^2$.
3. The position vector $\vec{OP} \coloneqq \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$.

The horizontal axis represents the **real axis** (Re), and the vertical axis represents the **imaginary axis** (Im).

1.2 Polar Form, Euler's Formula, and Exponential Representation

1.2.1 Polar Coordinates and Polar Form

Transforming the Cartesian coordinates (x, y) into plane polar coordinates (r, θ) via

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases} \tag{2}$$

allows z to be written in **trigonometric (polar) form**:

$$z = r(\cos\theta + \mathrm{i}\sin\theta). \tag{3}$$

1.2.2 Modulus (Absolute Value)

Definition 2 (Modulus)**.** The **modulus** $r \coloneqq |z|$ represents the Euclidean distance from the origin to $P(x, y)$:

$$|z| \coloneqq \sqrt{x^2 + y^2} \geq 0. \tag{4}$$

1.2.3 Argument and Principal Argument

The angle θ is defined as an **argument** of z, denoted by $\mathrm{Arg}(z)$. Since the trigonometric functions are 2π-periodic, $\mathrm{Arg}(z)$ is a multi-valued set:

$$\mathrm{Arg}(z) \coloneqq \arg(z) + 2k\pi, \quad k \in \mathbb{Z}. \tag{5}$$

The unique representative in the interval $(-\pi, \pi]$ is defined as the **principal argument**, denoted by $\arg(z) \in (-\pi, \pi]$:

$$\arg(z) \coloneqq \begin{cases} \arctan\left(\frac{y}{x}\right), & x > 0 \\ \arctan\left(\frac{y}{x}\right) + \pi, & x < 0,\ y \geq 0 \\ \arctan\left(\frac{y}{x}\right) - \pi, & x < 0,\ y < 0 \\ \frac{\pi}{2}, & x = 0,\ y > 0 \\ -\frac{\pi}{2}, & x = 0,\ y < 0 \end{cases} \tag{6}$$

(For $z = 0$, the modulus is $|0| = 0$, and the argument $\mathrm{Arg}(0)$ is undefined.)

1.2.4 Physical Meaning and Introduction of the Exponential Form

In physics and engineering (such as wave mechanics, alternating-current circuits, and quantum mechanics), complex exponentials are the natural language for describing **rotation**, **phase**, and **harmonic oscillation (phasors)**:

- **Rotation Operator:** The factor $\mathrm{e}^{\mathrm{i}\theta}$ acts as a unit rotator ($|\mathrm{e}^{\mathrm{i}\theta}| = 1$). Multiplying any vector by $\mathrm{e}^{\mathrm{i}\theta}$ preserves its magnitude while rotating it counterclockwise by an angle θ.
- **Harmonic Motion:** When the phase evolves uniformly with time, $\theta(t) = \omega t$, the term $\mathrm{e}^{\mathrm{i}\omega t}$ describes uniform circular motion on the complex plane. Its projection onto the real axis yields the standard harmonic oscillator $\cos(\omega t)$.

This bridge between circular rotation and trigonometry is codified as **Euler's Formula**:

$$\mathrm{e}^{\mathrm{i}\theta} \coloneqq \cos\theta + \mathrm{i}\sin\theta. \tag{7}$$

Substituting Euler's formula into the polar representation yields the **exponential form** of a complex number:

$$z \coloneqq r\mathrm{e}^{\mathrm{i}\theta}, \tag{8}$$

where $r = |z|$ represents the **amplitude** (magnitude) and θ represents the **phase** (orientation).

2 Operations and Geometric Transformations

2.1 Addition and Subtraction

For $z_1 = x_1 + \mathrm{i}y_1$ and $z_2 = x_2 + \mathrm{i}y_2$:

$$z_1 \pm z_2 \coloneqq (x_1 \pm x_2) + \mathrm{i}(y_1 \pm y_2). \tag{9}$$

Geometric Meaning: Addition and subtraction correspond to vector addition and subtraction of $\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$ and $\begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$ according to the parallelogram law or triangle rule.

2.2 Complex Conjugation

Definition 3 (Complex conjugate)**.** The **complex conjugate** of $z = x + \mathrm{i}y = r\mathrm{e}^{\mathrm{i}\theta}$ is defined by

$$\bar{z} \coloneqq x - \mathrm{i}y = r\mathrm{e}^{-\mathrm{i}\theta}. \tag{10}$$

Fundamental properties:

- $\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2$
- $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$
- $z\bar{z} = |z|^2 = x^2 + y^2$
- $\mathrm{Re}(z) = \frac{z+\bar{z}}{2}, \quad \mathrm{Im}(z) = \frac{z-\bar{z}}{2\mathrm{i}}$

2.3 Multiplication and Division

Let $z_1 = r_1\mathrm{e}^{\mathrm{i}\theta_1}$ and $z_2 = r_2\mathrm{e}^{\mathrm{i}\theta_2}$.

2.3.1 Multiplication

$$z_1 z_2 = (r_1 r_2)\mathrm{e}^{\mathrm{i}(\theta_1+\theta_2)}. \tag{11}$$

- **Modulus:** $|z_1 z_2| = |z_1||z_2|$
- **Argument:** $\mathrm{Arg}(z_1 z_2) = \mathrm{Arg}(z_1) + \mathrm{Arg}(z_2) \pmod{2\pi}$

Geometric Meaning: Multiplying a vector z by $z_1 = r_1\mathrm{e}^{\mathrm{i}\theta_1}$ dilates its length by factor r_1 and rotates it counterclockwise by an angle of θ_1.

2.3.2 Division

$$\frac{z_1}{z_2} \coloneqq \frac{z_1\bar{z}_2}{|z_2|^2} = \left(\frac{r_1}{r_2}\right)\mathrm{e}^{\mathrm{i}(\theta_1-\theta_2)}, \quad (z_2 \neq 0). \tag{12}$$

- **Modulus:** $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$
- **Argument:** $\mathrm{Arg}\left(\frac{z_1}{z_2}\right) = \mathrm{Arg}(z_1) - \mathrm{Arg}(z_2) \pmod{2\pi}$

Example 1 (Division by i)**.**

$$\frac{z}{\mathrm{i}} = \frac{z(-\mathrm{i})}{\mathrm{i}(-\mathrm{i})} = -\mathrm{i}z = z \cdot \mathrm{e}^{-\mathrm{i}\frac{\pi}{2}}. \tag{13}$$

Geometric Meaning: Multiplying by $-\mathrm{i}$ (or dividing by i) rotates the vector clockwise by $\frac{\pi}{2}$.

2.4 Powers, Binomial Expansion, and de Moivre's Theorem

For any positive integer $n \in \mathbb{N}^+$:

$$z^n = (r\mathrm{e}^{\mathrm{i}\theta})^n = r^n\mathrm{e}^{\mathrm{i}n\theta} = r^n(\cos(n\theta) + \mathrm{i}\sin(n\theta)). \tag{14}$$

Setting $r = 1$ gives **de Moivre's Theorem**:

$$(\cos\theta + \mathrm{i}\sin\theta)^n = \cos(n\theta) + \mathrm{i}\sin(n\theta). \tag{15}$$

In Cartesian form, z^n can be expanded using the binomial theorem:

$$z^n = (x + \mathrm{i}y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k}(\mathrm{i}y)^k, \tag{16}$$

where the **binomial coefficient** $\binom{n}{k}$ is defined by

$$\binom{n}{k} \coloneqq \frac{n!}{k!(n-k)!} = \frac{n(n-1)\cdots(n-k+1)}{k!}, \tag{17}$$

with $0! \coloneqq 1$.

2.5 Roots of Complex Numbers

If $w^n = z$ for $n \in \mathbb{N}^+$, w is an n-th root of z, denoted by $\sqrt[n]{z}$.

Let $z = re^{i\theta}$. There are exactly n distinct complex roots given by

$$w_k \coloneqq \sqrt[n]{r}\exp\left(i\frac{\theta+2k\pi}{n}\right) = \sqrt[n]{r}\left[\cos\left(\frac{\theta+2k\pi}{n}\right) + i\sin\left(\frac{\theta+2k\pi}{n}\right)\right] \tag{18}$$

for $k = 0, 1, 2, \ldots, n-1$.

2.5.1 Special Case: Roots of Unity ($\sqrt[n]{1}$)

When $z = 1 = 1 \cdot e^{i0}$, the solutions to $w^n = 1$ are

$$\omega_k \coloneqq e^{i\frac{2k\pi}{n}} = \cos\left(\frac{2k\pi}{n}\right) + i\sin\left(\frac{2k\pi}{n}\right), \quad k = 0, 1, \ldots, n-1. \tag{19}$$

Geometric Meaning: The n roots of unity form the vertices of a regular n-sided polygon inscribed in the unit circle $|w| = 1$, with one vertex located at $w = 1$.

3 Geometric Applications in the Complex Plane

3.1 Distance and Modulus Identities

The Euclidean distance between two points represented by z_1 and z_2 is

$$\operatorname{dist}(z_1, z_2) \coloneqq |z_1 - z_2|. \tag{20}$$

3.1.1 Key Identities and Inequalities

1. **Modulus–Conjugate Relation:**
$$|z|^2 = z\bar{z}. \tag{21}$$

2. **Parallelogram Identity:**
$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2). \tag{22}$$

3. **Modulus Expansion:**
$$|z_1 \pm z_2|^2 = |z_1|^2 + |z_2|^2 \pm 2\,\mathrm{Re}(z_1\bar{z}_2). \tag{23}$$

4. **Triangle Inequality:**
$$\big||z_1| - |z_2|\big| \le |z_1 \pm z_2| \le |z_1| + |z_2|. \tag{24}$$

3.2 Standard Curves and Lines

3.2.1 Circle

A circle C centered at z_0 with radius $r > 0$:

- **Implicit Form:**

$$|z - z_0| = r. \tag{25}$$

- **Parametric Form:**

$$z(t) \coloneqq z_0 + r\mathrm{e}^{\mathrm{i}t}, \quad t \in [0, 2\pi). \tag{26}$$

3.2.2 Straight Line

A straight line passing through two distinct points z_1 and z_2:

$$z(t) \coloneqq z_1 + t(z_2 - z_1), \quad t \in \mathbb{R}. \tag{27}$$

3.3 Angle Between Vectors and Angles in a Triangle

3.3.1 Directed Angle

The directed angle θ from vector $(z_3 - z_1)$ to vector $(z_2 - z_1)$ is

$$\theta \coloneqq \operatorname{Arg}\left(\frac{z_2 - z_1}{z_3 - z_1}\right). \tag{28}$$

3.3.2 Interior Angles of a Triangle

For a positively oriented (counterclockwise) triangle $\triangle ABC$ with vertices z_1, z_2, z_3:

$$\begin{cases} \angle z_1 \coloneqq \arg\left(\dfrac{z_3 - z_1}{z_2 - z_1}\right) \\ \angle z_2 \coloneqq \arg\left(\dfrac{z_1 - z_2}{z_3 - z_2}\right) \\ \angle z_3 \coloneqq \arg\left(\dfrac{z_2 - z_3}{z_1 - z_3}\right) \end{cases} \tag{29}$$

Theorem 1 (Sum of interior angles)**.** *The sum of the interior angles of a nondegenerate triangle equals π.*

Proof. Consider the cyclic product

$$P \coloneqq \left(\frac{z_3 - z_1}{z_2 - z_1}\right) \cdot \left(\frac{z_1 - z_2}{z_3 - z_2}\right) \cdot \left(\frac{z_2 - z_3}{z_1 - z_3}\right) = -1. \tag{30}$$

Taking arguments on both sides,

$$\operatorname{Arg}(P) = \operatorname{Arg}\left(\frac{z_3 - z_1}{z_2 - z_1}\right) + \operatorname{Arg}\left(\frac{z_1 - z_2}{z_3 - z_2}\right) + \operatorname{Arg}\left(\frac{z_2 - z_3}{z_1 - z_3}\right) = \operatorname{Arg}(-1) = \pi + 2k\pi, \quad k \in \mathbb{Z}. \tag{31}$$

For interior angles of a nondegenerate triangle, each angle satisfies $0 < \angle z_i < \pi$, so

$$0 < \angle z_1 + \angle z_2 + \angle z_3 < 3\pi. \tag{32}$$

Thus, k must be 0, which proves

$$\angle z_1 + \angle z_2 + \angle z_3 = \pi. \tag{33}$$

□

3.4 Geometric Characterization of Circles (Thales's Theorem)

Theorem 2 (Thales's Theorem)**.** *A point z lies on the circle with diameter endpoints z_1 and z_2 (with $z \neq z_1$) if and only if*

$$\operatorname{Re}\left(\frac{z_2 - z_1}{z - z_1}\right) = 1. \tag{34}$$

Proof. Rearranging the algebraic fraction,

$$\frac{z_2 - z_1}{z - z_1} - 1 = \frac{(z_2 - z_1) - (z - z_1)}{z - z_1} = \frac{z_2 - z}{z - z_1}. \tag{35}$$

Taking the real part of both sides yields

$$\operatorname{Re}\left(\frac{z_2 - z}{z - z_1}\right) = 0 \iff \frac{z_2 - z}{z - z_1} \text{ is purely imaginary.} \tag{36}$$

This is equivalent to

$$\arg\left(\frac{z_2 - z}{z - z_1}\right) = \pm\frac{\pi}{2} \iff \angle(z_1, z, z_2) = \frac{\pi}{2}. \tag{37}$$

By Thales's Theorem, the locus of points subtending a right angle across segment $[z_1, z_2]$ is the circle with diameter $[z_1, z_2]$. This also satisfies the Pythagorean theorem:

$$|z_1 - z_2|^2 = |z - z_1|^2 + |z - z_2|^2. \tag{38}$$

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